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🔗 GitHub: github.com/Xuan-yi-yan | vLLM Contributor | DeepSeek-V3 Community | DeepSpec RFC #52

Python PyTorch vLLM Contributor KV Cache Hybrid Model Alignment Prefix Caching White-Box Architecture
Cross-Attention RMSNorm Gating System Hierarchical Adam Gradient Diagnostics NaN Rollback Multi-Agent
LLM API Integration Mixed Precision Multimodal Unified Architecture Spherical Normalization ABCDEF Cognitive Stack
Speculative Decoding Claude/Gemini Benchmark

1. vLLM / vLLM-Ascend Open-Source Contributions (2026.07, 85K stars)

The world's largest LLM inference engine + Huawei Ascend ecosystem | 4 PRs/RFCs | Verified on RTX 5070

- **vLLM PR #47491**: Fixed hybrid model Mamba group zero-hit prefix cache truncation. 5-line fix, 9 reviewers (incl. WoosukKwon, vLLM founder)
- **vLLM-Ascend PR #11314**: Progressive alignment algorithm fix for DeepSeek V4 MTP 32K→16K cache truncation. `_progressive_align_steps` + early-exit guard, CI all green
- **vLLM-Ascend RFC #11356**: PD-disaggregated Mamba state independent channel routing architecture proposal. HybridTransferTracker control/data plane decoupling + `min()` graceful degradation + Feature Flag
- **vLLM RFC #45702**: DAG + CoW solution for Mamba concurrent prefix sharing. No CUDA kernel modifications required. Physically verified on RTX 5070
- **vLLM #43090 Root-Cause Analysis**: Traced granite hybrid model prefix-caching output divergence — a correctness bug undetected by core maintainer njhill. 4-link evidence chain pinpointing `has_initial_states` misjudgment. Published independent repro script (gist) + transparently disclosed own PR boundary
- **DeepSpec RFC #52**: Submitted V22 white-box architecture reference document to DeepSeek's official speculative decoding codebase — spherical normalization, 3-tier semantic cache, unified multimodal token as architectural reference for DSpark. 5 open adoption hooks for community

2. DeepSeek-V3 Community Technical Contributions (2026.06, 104K stars)

Officially featured in June Monthly Community Report — "Researchers Provided Key Diagnoses and Solutions" section

- **Graviton Anchors**: Concept adopted as formal terminology by AmoebaFPS independent research lab, formalized as GDI mathematical framework + PyTorch MVS implementation
- **Attention Lock-in Diagnosis**: #1125 identified inference-time suffix constraints causing meta-signal contamination of content-signal. Root cause adopted as diagnostic paradigm by community (#1463 citation)
- **Safety Anchor Injection**: #1368 proposed "Safety is a gate, not a weight" — three inference-time injection schemes (Anchor Injection / Logit Bias / CoT Check)
- **Semantic Pre-annotation Middleware**: #1420 proposed KV-pool semantic indexing. Validated in production by HeartFlow system (3 months data, 40% token reduction)

- **AmoebaFPS Public Response:** "Most significant insight received... parameter-space bottleneck is the missing piece." GDI framework adopted to formalize my Graviton Anchors concept
- **Five issue root-cause analyses:** #5083(Qwen) + #1368 + #1420 + #1125 + #1460, spanning five dimensions (OS→Security→Attention→Reasoning→Training)

3. V21→V22 White-Box Multimodal Cognitive Engine (Independent Development, 2026.05-07)

Extended from V21 single-modality text engine to V22 three-modality (Text/Vision/Audio) unified architecture — a single ABCDEF cognitive stack handles three modalities, with signals unified as "integer/frequency → Spherical Embedding + Attributes", rejecting per-modality siloed models (Whisper/ViT). 12+ modules, ~9.9M parameters, fully interpretable. Trained on single RTX 5070 12GB. The underlying layered memory management, gradient diagnostics, and NaN protection mechanisms are directly transferable to large-scale distributed training and inference engine stability design.

Architecture

- **CharEmbed:** ~3000 chars × 256D, orthogonal initialization — each character maps to an independent direction
- **P3 AttributeStack:** Zero-parameter rule engine (14 sub-modules), 128D × 126 slots across 3 hierarchical layers (word-class/semantics/syntax/emotion/logic/modifier/domain/style)
- **P7 CrossSentence:** 16-head × 16D, Q=A tokens, K/V=full vocabulary. **RMSNorm Q/K anti-explosion**, learnable temperature
- **P3-L AttributeLinkage:** 312 heads / 23 groups, independent per-group attention (anti-contamination) with cross-modal groups
- **ABC Tri-Stage Logic Review:** Stage A (20-class structural decision) → Stage B (384D content) → Stage C (48D tone). Reviews attribute coherence
- **Gate (3-channel parallel):** base/abc/sent signals each 128D independent channel → Explore → Meta → Gate[256] modulates P6
- **P6 Tied-Weight Decoder:** 128-head × Linear(256,256), tied weights — parameters reduced from ~40M to 8.6M
- **ABC' Pressurization (V21.1):** Post-P3-L pressurization module (128→192→128), additive injection into P6, cold-start safe (gain=0.01)

V22 Multimodal Extension (2026.07, 3-Modality Milestone)

- **Unified Token Engine (extension, not rewrite):** Text (char_embed), Vision (VisualEmbed, 16×16 raw RGB → 256D spherical), Audio (AudioEmbed, raw FFT spectrum → 256D spherical) — unified as "256D spherical Embed + Attributes" isomorphic tokens. Single P7→ABC→P6 stack handles all three modalities. No modality-specific branches.
- **ABCDEF Hexa-Cognitive Stack:** Extended ABC with D (Relation Review: relations ARE P3 attributes, ABC reviews coherence, no GNN) + E (Memory Read-Write Head: sent_vec cosine retrieval + P3 attribute Jaccard hybrid scoring + file-persisted 3-tier cache). Attribute dimension: 128→384 (language 128 + vision 256)
- **Spherical Normalization Root-Cause Fix:** Diagnosed crash chain: "numerical range overflow → gradient shock → sigmoid saturation → gate gradient death → full-network collapse." Root-cured via F.normalize unit-sphere constraint — cold-start gain reduction only delayed the inevitable.
- **Cross-Modal Contrastive Loss Directionality:** Image/Audio→Text uses cosine similarity. Text→Image/Audio uses Pearson correlation (requires numerical linear correspondence)

- **3-Modality Verification:** Text: loss→0.0001 auto-stop. Vision: "image(apple)→predict:苹果" correct. Audio: "simulated FFT→text" data flowing. All through a single engine.

Training System

- **7 Independent Adam Optimizers + Hierarchical Learning Rates:** Precisely calculated from gradient/parameter ratios — P7×0.05 (97% suppression), Gate×5.0 (5x boost), ABC×3.0
- **5-Layer Intelligent Safeguard:** ① ac_batch real-time detection ② CE single-spike clamp ③ aux_scale progressive self-heal (5%/step) ④ NaN auto-rollback (from best.pt) ⑤ loss<0.0001 auto-stop
- **Gradient Diagnostic System:** Per-module grad/param ratio real-time tracking
- **Data Engineering:** Greedy coverage sampling (1500 pairs = 89% vocabulary), dirty data filtering (4360 pairs / 8.3%), 52K full analysis

Bug Hunter

- 25+ hypothesis tests → 18 disproved → 12 confirmed real bugs → **Discovered "Fake 2.1%": eval script dimension mismatch**
- **Sigmoid bias dead gradient:** bias=2.0 → gate path dead in 5 epochs, 6-hour root-cause chase
- 36-hour "Operation Dawn": 9 rounds of dual-AI debate + 60 code modifications + 4 architecture upgrades

4. Dual-Agent Debate Audit System (2026.06)

- **Architecture:** DeepSeek (read-only analysis) + Qwen (read-write review), auto debate loop (max 5 rounds), convergence detection. Applied to V22 architecture review: Claude-Opus-4.5 vs Gemini-2.5-Pro cross-model debate.
- **7 Tools:** read_file / search_files / search_content / list_dir / web_fetch / run_cmd / write_file
- **Permission Control:** DeepSeek readonly (write/run_cmd hidden), Qwen full permissions. JSON+XML dual-format tool-call parsing
- **Track Record:** 9 debate rounds located 6 P0 bugs, including eval script dimension mismatch (accuracy 2.1%→78.6% truth)

5. Black-Box V2 Multi-Tuple Architecture (2026.06, 3h from zero to validation)

- Trainable N-gram character tuples replacing rule engine — "which characters frequently appear together" learned by the model
- 10M parameters, 1500 pairs CE normal descent (6→2.2), architecture validation passed

Core Skills

White-Box Architecture from Zero vLLM/KV Cache Hybrid Model Alignment Attention Mechanism Gradient Diagnostics
Bug Hunter (25 hypotheses) Hierarchical Optimizers NaN Rollback Multi-Agent Systems Full-Stack Independent Dev
Technical Writing (220+ docs) Speculative Decoding